Variance-Reduced Q-Learning over Static and Time-Varying Networks

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Abstract

This paper studies decentralized multi-agent reinforcement learning (MARL) for discounted tabular MDPs, where N agents interact with the same environment but can communicate only over a graph. The goal is for *every* agent to learn the optimal action-value function $Q^\star$ efficiently, achieving the same statistical benefit as centralized learning with NT total samples. The paper proposes `VRDQ` a.k.a Variance-Reduced Diffused Q-learning, an epoch-based method that couples (i) variance reduction via epoch-wise empirical Bellman operators built from batched samples, and (ii) limited diffusion/consensus for only L rounds per epoch. The main theorems show that with appropriate choices of epoch length and diffusion steps, all agents attain a high-probability bound of the form $\widetilde{\mathcal{O}}(1/\sqrt{NT})$, and the guarantee extends to a standard model of undirected time-varying networks. Empirical results on a tabular gridworld illustrate the expected $1/\sqrt{N}$ collaboration gain and show that once diffusion is adequately tuned, performance becomes largely insensitive to the specific network topology.

Theoretical Results

`VRDQ` couples epoch-wise variance reduction with L rounds of diffusion per epoch, enabling every agent to learn $Q^\star$ efficiently over static and time-varying networks.

- **Collaboration Gain.**
  With high probability,

$$\frac{1}{N}\sum_{i=1}^{N}\|Q_{i,K}-Q^\star\|_\infty \;\le\; \widetilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}}\right),$$

  so the error matches the centralized statistical scaling corresponding to NT total samples across N agents.

- **$\tilde{\mathcal{O}}(1)$ communication.** Agents perform only L consensus steps per epoch, with $L=\tilde{\mathcal{O}}(1)$ so the number of communication rounds is

$$K=\mathcal{O}(\log(NT))\,,$$

  rather than $\Theta(T)$.

Experiments

Figure 1. Plots of the ℓ_∞ error $E_K = \sum_{i\in[N]}\|Q_{i,K}-Q^*\|_\infty/N$ for `VRDQ` as a function of the number of epochs K, with varying number of agents.

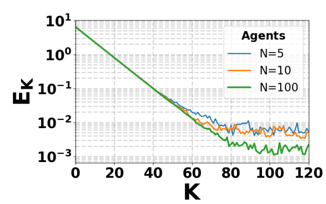
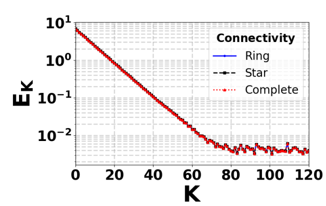


Figure 2. Plots of E_K for fixed N under different network topologies e.g., ring/star/complete, illustrating that with sufficient diffusion steps L, performance is largely insensitive to the graph structure.



Introduction

In decentralized MARL, multiple agents learn from local interactions while exchanging messages with neighbors in a communication graph. Unlike federated learning with a central server, decentralized settings are relevant when a server is unavailable or undesirable (e.g., swarms, ad-hoc networks, peer-to-peer systems). A central question is whether decentralized algorithms can match the statistical efficiency of centralized learning, and what communication burden is required.

Classical distributed Q-learning is known to converge under various assumptions, but finite-time guarantees reveal two bottlenecks:

- **Noisy Bellman targets:** standard Q-learning uses single-sample temporal-difference targets, which have high variance.
- **Imperfect consensus:** decentralized averaging introduces additional error unless sufficient communication rounds are performed.

The paper addresses these by building a *variance-reduced* learning signal via epoching and by performing only a *limited* number of consensus steps per epoch. It further studies not only static graphs but also practically relevant *time-varying* networks.

Main Results

■ **Objective.** There are N agents connected by an undirected graph $\mathcal{G}$ or a sequence of graphs $\{\mathcal{G}(t)\}$ in the time-varying case. All agents face the same discounted tabular MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$. Each agent i maintains its own Q-iterate $Q_{i,k}$ at epoch $k \in [K]$, and the aim is that all agents achieve small error

$$e_{i,K} := \|Q_{i,K} - Q^\star\|_\infty \quad \text{after} \quad T = KH \text{ samples per agent.}$$

■ **Algorithm: `VRDQ`.** Time is partitioned into K epochs of length H. Each epoch consists of:

(1) Local empirical Bellman operator (variance reduction). Agent i uses its H batched samples in epoch k to estimate the transition kernel $\mathcal{P}(\cdot|s,a)$ for every (s,a), yielding an empirical model $\widehat{\mathcal{P}}_{i,k}$. This defines an empirical Bellman backup operator

$$(\widehat{\mathcal{T}}_{i,k} f)(s,a) \;=\; \mathcal{R}(s,a) \;+\; \gamma \sum_{s'} \widehat{\mathcal{P}}_{i,k}(s'|s,a) \max_{a'} f(s',a').$$

The message to be diffused is a variance-reduced backup of the current iterate, e.g., $d_{i,k}^{(0)} := \widehat{\mathcal{T}}_{i,k-1} Q_{i,k}$.

(2) Diffusion/consensus for L rounds (network averaging). Agents apply L iterations of consensus using a mixing matrix W (static graph) or time-varying matrices $W(t)$:

$$d_{i,k}^{(\ell+1)}(s,a) \;=\; \sum_{j \in \mathcal{N}_i} w_{ij}\, d_{j,k}^{(\ell)}(s,a), \qquad \ell = 0, 1, \ldots, L-1.$$

After L rounds, $d_{i,k}^{(L)}$ approximates the network average of the initial messages, with an error controlled by the graph mixing rate.

(3) Local Q-update. Each agent updates:

$$Q_{i,k+1}(s,a) \;=\; (1-\alpha)\, Q_{i,k}(s,a) \;+\; \alpha\, d_{i,k}^{(L)}(s,a).$$

Thus, communication is *epoch-level*: only L mixing steps per epoch, not per environment interaction.

■ **Static networks and Finite-time guarantees.** Let $\rho \in (0,1)$ denote a diffusion factor governing consensus mixing speed. With suitable choices of epoch count K, step size α, and diffusion budget L chosen as a logarithmic function of the horizon and network parameters, the paper proves that with high probability, *every* agent satisfies a bound of the form

$$\|Q_{i,K} - Q^\star\|_\infty \;\le\; \widetilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}}\right).$$

■ **Interpretation.**

- **Centralized-rate learning:** the $\widetilde{\mathcal{O}}(1/\sqrt{NT})$ term matches the statistical rate one would expect if NT total samples were pooled centrally, demonstrating a $1/\sqrt{N}$ collaboration gain.
- **Communication–accuracy tradeoff:** the required L grows only logarithmically with the problem horizon but depends inversely on the network mixing strength, meaning sparse graphs may need more diffusion rounds.
- **Role of variance reduction:** using empirical Bellman operators based on H samples per epoch reduces the noise in the diffused messages, making the overall recursion stable and allowing fewer consensus rounds compared to diffusing raw targets.

■ **Extension to Time-Varying Networks.** The analysis extends to undirected time-varying graphs under a standard joint-connectivity/mixing condition: over windows of length B, products of the mixing matrices contract disagreements by a factor $\omega \in (0,1)$. Under an appropriate diffusion budget $\bar{L}$ that depends on (B,ω), the same qualitative bound persists:

$$\max_i \|Q_{i,K} - Q^\star\|_\infty \;\le\; \widetilde{\mathcal{O}}\left(\frac{1}{\sqrt{NT}}\right).$$

Experimental Results

Metric. The paper reports the average ℓ_∞ error across agents:

$$E_K \;=\; \frac{1}{N}\sum_{i=1}^{N} \|Q_{i,K} - Q^\star\|_\infty,$$

plotted versus the number of epochs K.

Scaling with number of agents. Keeping other parameters fixed and varying N, the curves show that larger N leads to a smaller final error level, consistent with the predicted $1/\sqrt{N}$ collaboration gain.

Impact of connectivity/topology. For a fixed number of agents (e.g., $N = 100$), the paper compares different network structures such as a ring, a star, and a complete graph. A separate consensus-gap diagnostic indicates that these topologies mix at different speeds, but once the diffusion budget L is chosen sufficiently large, the learning curves E_K become similar across topologies. This supports the theory: topology primarily affects how large L must be to make consensus error negligible; after tuning, the dominant term is the statistical error $\widetilde{\mathcal{O}}(1/\sqrt{NT})$.

Takeaway. The experiments corroborate that `VRDQ` can achieve centralized-like learning accuracy in decentralized networks with a modest number of diffusion steps per epoch, and that the method behaves robustly across static and time-varying connectivity patterns when consensus is properly controlled.